

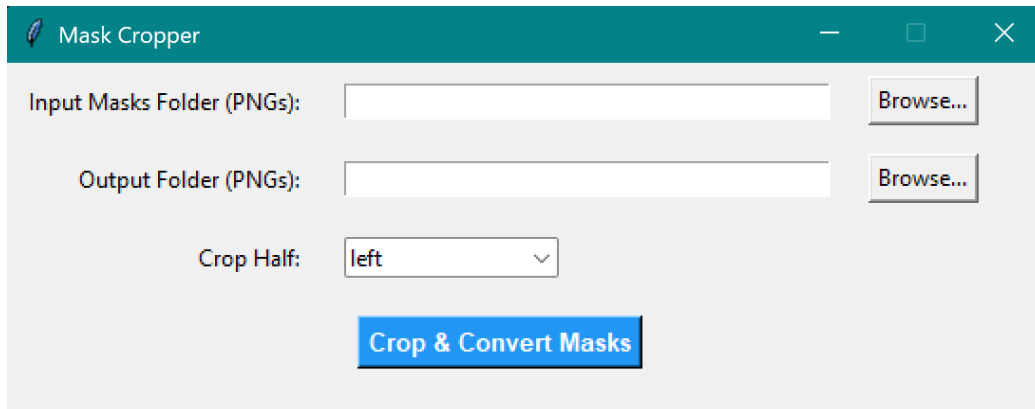
Processing pipeline: scripts tutorials

1. Correct the masks

Using SAM3GUI or the Mask Annotator app.

2. Crop the masks into separate folders – one for each channel

Run the script by typing `python croppingmasks.py` in the terminal.



This GUI will appear.

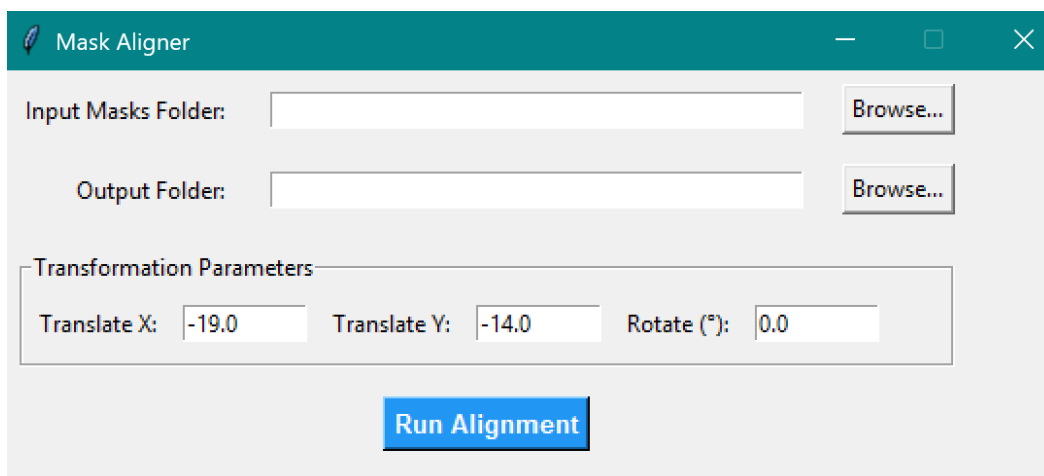
Select the folder containing the full masks in **Input Masks Folder** and keep the right half for the neuron masks (green channel) and the left half for the worm masks (red channel).

3. Calculate alignment and ROI

With the CaTracker pipeline. Align and track one frame and write down the ROI.

4. Apply this alignment to the green channel masks (the neuron masks)

Run the script by typing `python alignmasks.py` in the terminal.

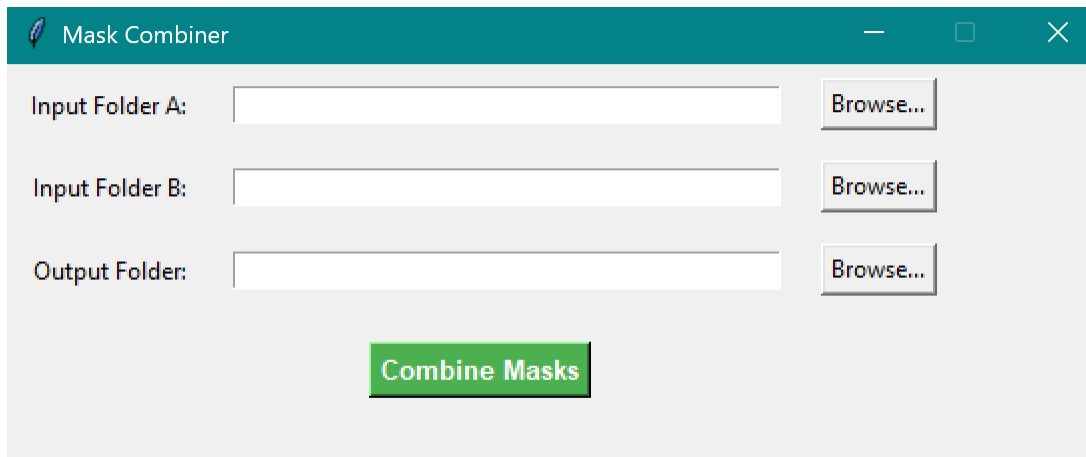


This GUI will appear. Select the Input Masks Folder to be the green channel masks and the output folder to be green channel masks aligned. Then fill in the alignment you wrote down in step 3 but change the sign.

So, if you wrote down dx: -3 and dy: -5, fill in 3 and 5 respectively.

5. Merge the red channel and green channel masks together

Run the script by typing `python merge-masks.py` in the terminal.



This GUI will appear.

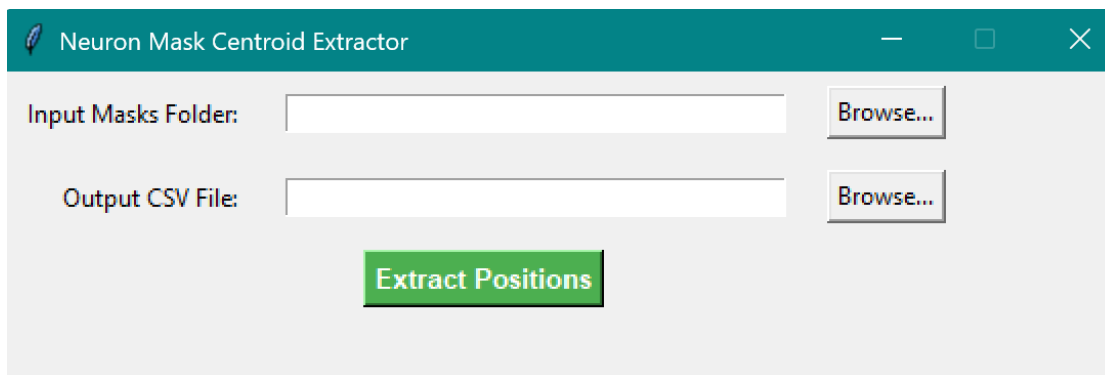
Select the red and green channels masks folders for Input Folder A and Input Folder B.

Select a new Output Folder to save the merged masks to.

Combine the masks.

6. Extract the neuron positions from the aligned green masks (see step 4)

Run the script by typing `python extract_neuron_positions.py` in the terminal.

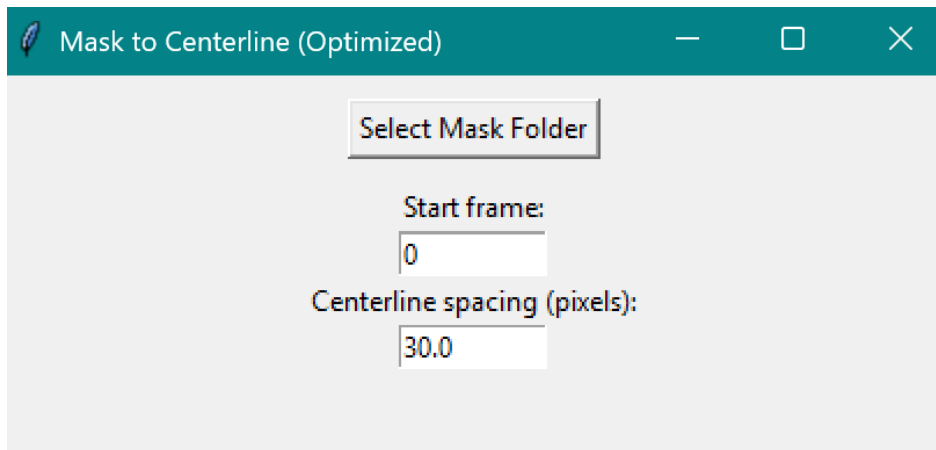


This GUI will appear.

Input the aligned neuron masks as the Input Masks Folder and the csv will be automatically saved in the root folder as the name you define.

7. Extract the centerlines from the merged red channel masks of step #5

Run the script by typing `python masktocenterline.py` in the terminal.

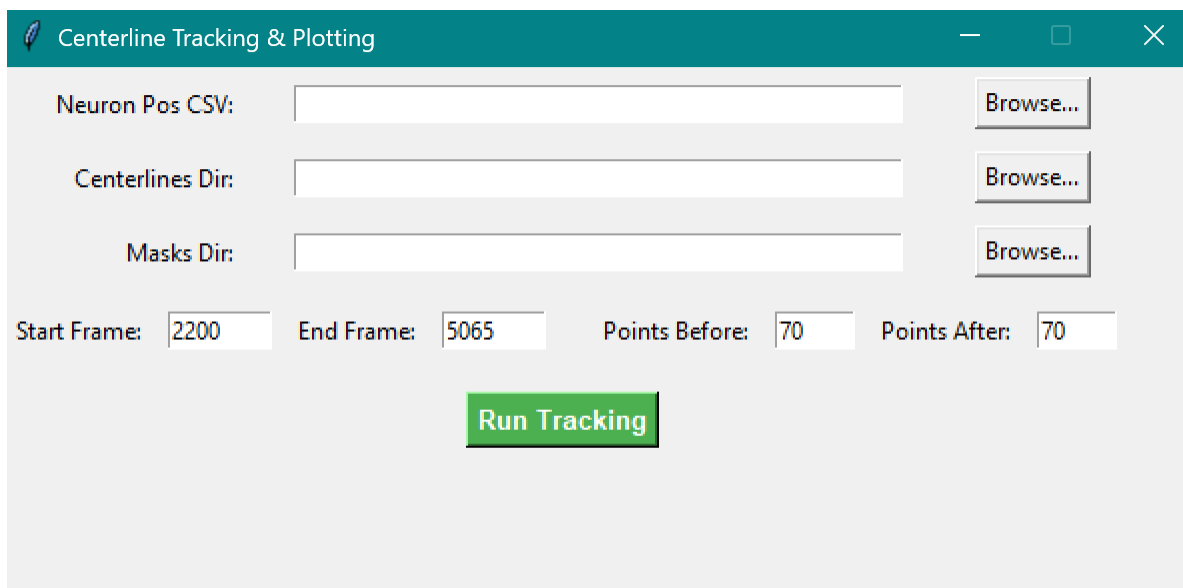


Select the merged masks of step #5 as the Mask Folder and keep the default centerline spacing of 30 pixels.

A folder called 'centerlines' will appear in the same folder as the masks.

8. Align the centerlines with the neuron positions from step #6

Run the script by typing `python track_fixed_pts_neuron.py` in the terminal



This GUI will appear.

The Neuron Pos CSV is the CSV saved in step #6.

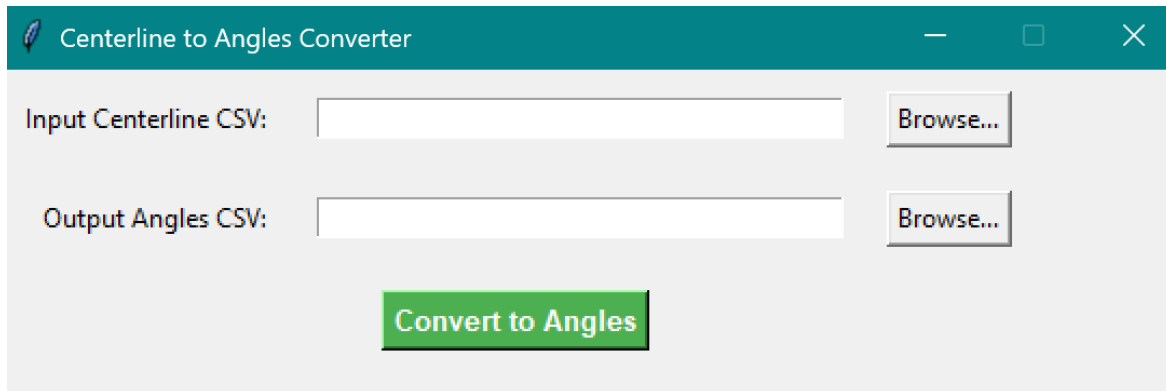
The Centerlines Dir is the folder containing all the centerlines from step #7.

The Masks Dir is the folder containing all the merged masks from step #5.

The resulting CSV with all matched points will be titled 'matched_points.csv' and will be saved in the same folder as the masks and the centerlines, along with a new folder titled 'Overplotted matched points' with png for verification.

9. Calculate the angles from the matched points

Run the script by typing `python angles.csv` in the terminal.

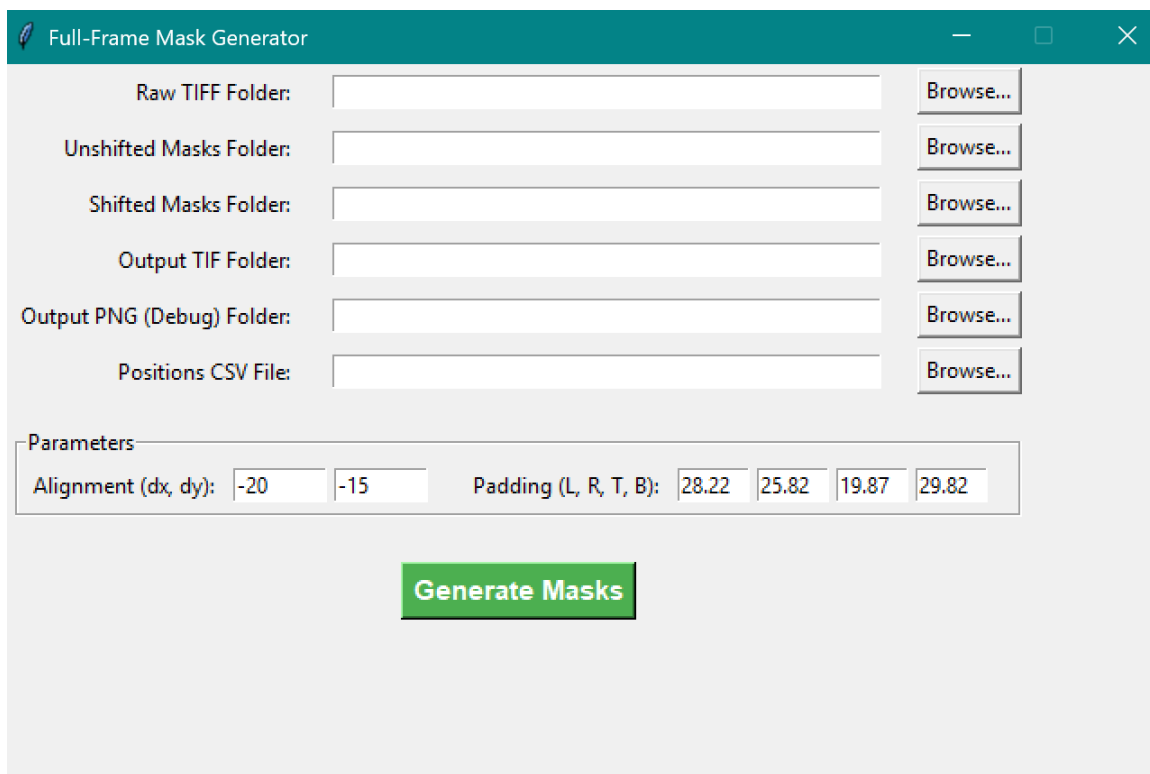


A screenshot of a software window titled "Centerline to Angles Converter". The window has a teal header bar with a feather icon, a minus button, a maximize button, and a close button. The main area is light gray and contains two input fields with "Browse..." buttons: "Input Centerline CSV:" and "Output Angles CSV:". Below these is a large green button labeled "Convert to Angles".

For the Input Centerline CSV, select the 'matched_points.csv' obtained in step 8.

10. Create full masks for the raw TIFF files

Run the script by typing `python tiffmasks.py` in the terminal.



A screenshot of a software window titled "Full-Frame Mask Generator". The window has a teal header bar with a feather icon, a minus button, a maximize button, and a close button. The main area is light gray and contains six input fields with "Browse..." buttons: "Raw TIFF Folder:", "Unshifted Masks Folder:", "Shifted Masks Folder:", "Output TIF Folder:", "Output PNG (Debug) Folder:", and "Positions CSV File:". Below these is a "Parameters" section with two groups of input fields: "Alignment (dx, dy):" with values "-20" and "-15", and "Padding (L, R, T, B):" with values "28.22", "25.82", "19.87", and "29.82". At the bottom is a large green button labeled "Generate Masks".

The Unshifted Masks Folder should be the masks of the worm merged in step #5.

The Positions CSV File should be the neurons positions from step #6.

The padding are the ROI positions.

11. Extract the signal from the raw TIFF files using the full masks

Run the script by typing `python tiffsignal.py` in the terminal.

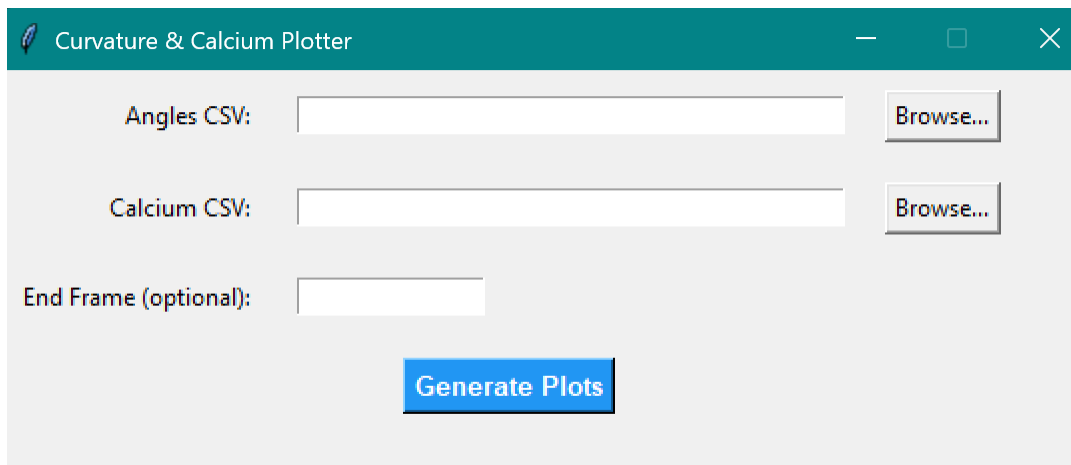


The screenshot shows a window titled "TIFF Signal Extractor" with a teal header bar. Inside the window, there are three input fields for folder selection: "Raw TIFF Folder:", "Mask TIFF Folder:", and "Output Folder:". Each field is accompanied by a "Browse..." button. At the bottom center of the window is a green button labeled "Extract Signals".

Two CSV files will be saved: one containing the total intensity of all pixels in the ROI and another one containing the 95% top brightest pixels.

12. Plot the curvature and signal

Run the script by typing `python all_plots.py` in the terminal.



The screenshot shows a window titled "Curvature & Calcium Plotter" with a teal header bar. Inside the window, there are three input fields: "Angles CSV:", "Calcium CSV:", and "End Frame (optional):". Each of the first two fields has a "Browse..." button next to it. At the bottom center of the window is a blue button labeled "Generate Plots".

Select the Angles CSV saved in step #9.

Select the Calcium CSV saved in step #11.

Then generate the plots and the plot will appear in a separate window.